

Viewfinder tech

According to rumours Olympus is working on a mirrorless camera that features an innovative viewfinder

Prime lens patents

Canon files patents for these lenses: 50mm f/1.4, 85mm f/1.2, 85mm f/1.8, 100mm f/2.0, 135mm f/2.0, and 200mm f/2.0

the mode dial lies the four way pad which also has a scrolling mechanism surrounded by the display, menu, video record and preview mode buttons on the four edges. The HS in the naming convention indicates that the camera belongs to the HS System which stands for high sensitivity. Both the cameras sport a backside illuminated CMOS sensor which enables better low light performance as compared to the CCD sensor which was seen in its predecessors. One design flaw is the pop-up flash unit. It pops up everytime you fire up the camera, and when you are using the flash, there is no place to keep your other hand's index finger making shooting with flash cumbersome. As far as the features are concerned, both the cameras sport pretty much most of the features seen on the S100 including slow motion video capturing, full HD resolution video shooting and a range of filters, etc.

The DMC TZ20 was the only camera we got from Panasonic in the entire test. It comes in a plastic body and is quite comfortable to hold thanks to its comparatively lighter weight. At 16x optical zoom, the DMC TZ20 had the highest optical zoom in the Travel Zoom category. A GPS receiver is housed within the camera. Notches were rarely seen on any camera in the test, but the DMC TZ20 has two of them – one on the top for power on/off and the other on the rear side to switch between the camera and preview modes.

Adjusting the aperture and shutter involve pressing the exposure button to activate respective icons on the screen. The four way pad does not scroll like the one seen on Canon. The Q Menu button brings up contextual menus in every mode. It has a full manual mode and also has a 3D mode which allows you

to shoot 3D images, which can be viewed only on a 3D TV or 3D monitor. Its 461k dot screen is touch sensitive and you can activate the focus tracking by touching an area on the screen which shows your subject. You can also fire a shot by touching the LCD screen.

The Nikon Coolpix S8200 is another camera that houses a backside illuminated CMOS sensor. The model we got was a metallic red colored one with a soft rubbery finish on the front face giving excellent grip. The S8200 has 14x optical zoom, but you do not have any options for manually controlling the camera. One look at the mode dial and you know this is camera meant for entry level users. You can go in the Menu and can further customise each mode, but there isn't much to adjust for the enthusiast. We liked the in-camera HDR mode which clicks a series of images and then stitches them into one with the final shot having a high dynamic range. The night landscape mode takes a series of five consecutive shots at high shutter speed and combines it to form a single image.

Performance

Under the ISO testing we noticed a prominent difference among the cameras. Lets starts with the TZ20 – upto ISO 400 the overall image quality is decent and at ISO 400 we can see noise only at a 100 percent crop. ISO 800 onwards the noise starts to ruin the fine details in the image. The Nikon S8200 showed great details upto ISO 400, at ISO 800 the noise starts creeping in but the image is not as bad as was observed with the TZ20 at ISO 800. At ISO 1600 you start losing the detailing and noise is quite prominent. With the Canon cameras, the image

quality was quite good upto ISO 800. At ISO 1600 some noise does start to show, but we still got the details on the threadbox and deo bottles and the larger text on the magazine. On the whole the ISO performance of the SX220HS was slightly better than the SX230HS.

In our sharpness and detail test, we noticed that all cameras except the TZ20 gave great detail at ISO 200 displaying each individual thread on the threadbox and legible



Canon Powershot SX 230HS

PCB text. It's at ISO 400 that we start to notice slight differences. With the TZ20, minute detail was lost at higher ISOs whereas with the Nikon S8200 the IC numbers started to get a bit blurry. The Canon SX220HS was the only camera which did not show much deviation in detailing at ISO 400 thanks to its better higher ISO performance. Images from both the SX220HS and SX230HS were sharp at ISO 400.

Under the low light shooting mode, we found that the output from the in-camera low light modes was not up to the mark. For instance, in the Canon cameras, at ISO 400 in Program mode, you get more details in the image under low light than the camera's own low light mode

, which tends to set the ISO to values over 1600, which leaves you with little detail. With the Nikon S8200, we found that the Night Landscape mode was pretty decent to shoot in low light. The Panasonic TZ20 was not very good under low lights. Even the photographs taken in Night mode, lacked details.

Flash testing is where the so-far invincible Canons tasted some defeat. The Nikon S8200 provided the best overall flash output, the words on

the laptop screen were readable, the lighting in the overall scene was balanced and there was hardly any vignetting. With the Canon cameras, we noticed that the legibility of the laptop screen was not that great and the overall lighting did not seem balanced. The TZ20 tended to blow up the lit screen of the laptop but it lit up the overall scene well.

As far as video performance went, the Canon cameras were much better than the Nikon and Panasonic ones. In the indoor shooting test, noise was not suppressed very well, but focussing wasn't an issue with the Canon SX220HS and SX230HS. We found the Panasonic TZ20 to be hunting for focus at the telephoto end. Same was the case with the Nikon S8200 despite keeping the